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## Coupled-inductor-based power-device circuit topology


### Abstract

According to an aspect of the present disclosure, an uninterruptible power supply (UPS) system is provided. The UPS system includes a first input configured to be coupled to an input power source, a second input configured to be coupled to an energy storage device, an output configured to provide output power, a power conversion circuit (PCC) configured to convert power received from at least one of the input power source or the energy storage device, an output circuit coupled to the PCC and the output, and a controller. The PCC includes a first inductor and a second inductor magnetically coupled to the first inductor. The controller is configured to control the PCC to provide, via the first inductor, DC power derived from the input power source to the output circuit, and provide, via the first and second inductors, DC power derived from the energy storage device to the output circuit.



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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. patent application Ser. No. 16/988,215, titled COUPLED-INDUCTOR-BASED POWER-DEVICE CIRCUIT TOPOLOGY, filed on Aug. 7, 2020, which claims priority under 35 U.S.C. § 119 to Indian Patent Application No. 201911032134, titled COUPLED-INDUCTOR-BASED POWER-DEVICE CIRCUIT TOPOLOGY, filed on Aug. 8, 2019. Each application referenced above is hereby incorporated by reference in its entirety for all purposes.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) At least one example in accordance with the present invention relates generally to power devices.

2. Discussion of Related Art

(2) The use of power devices, such as UPSs, to provide regulated, uninterrupted power for sensitive and/or critical loads, such as computer systems and other data processing systems, is known. Known uninterruptible power supplies include on-line UPSs, offline UPSs, line interactive UPSs, as well as others. Online UPSs provide conditioned AC power as well as back-up AC power upon interruption of a primary source of AC power. Offline UPSs typically do not provide conditioning of input AC power, but do provide back-up AC power upon interruption of the primary AC power source. Line interactive UPSs are similar to off-line UPSs in that they switch to battery power when a blackout occurs but also typically include a multi-tap transformer for regulating the output voltage provided by the UPS. Certain UPSs can include multiple power conversion stages.

SUMMARY

(3) According to at least one aspect of the present disclosure an uninterruptible power supply (UPS) system including a first input configured to be coupled to an input power source, a second input configured to be coupled to an energy storage device, an output configured to provide output power, a power conversion circuit configured to convert power received from at least one of the input power source or the energy storage device, the power conversion circuit including a primary branch portion having a first inductor, and a backup branch portion having a second inductor, the second inductor being magnetically coupled to the first inductor, an output circuit coupled to the power conversion circuit and to the output, and a controller coupled to the power conversion circuit and to the output circuit, and configured to control the power conversion circuit to provide, in a normal mode of operation via the first inductor, DC power derived from the input power source to the output circuit, and control the power conversion circuit to provide, in a backup mode of operation via the first inductor and the second inductor, DC power derived from the energy storage device to the output circuit.

(4) In an embodiment, the output circuit includes a first capacitor configured to be coupled to the power conversion circuit and a second capacitor configured to be coupled to the power conversion circuit. In one embodiment, the primary branch portion includes a third inductor, and the backup branch portion includes a fourth inductor, and wherein the third inductor is magnetically coupled to the fourth inductor. In at least one embodiment, the second inductor is coupled to a first switching device, and wherein the fourth inductor is coupled to a second switching device. In some embodiments, the controller is further configured to control the first switching device to charge a first capacitor, and is configured to control the second switching device to charge a second capacitor.

(5) In at least one embodiment, the controller is further configured to control the first switching device to charge the first capacitor at a first rate and is configured to control the second switching device to charge the second capacitor at a second rate different than the first rate. In some embodiments, the controller is further configured to control the first switching device and the second switching device to charge the first capacitor simultaneously with the second capacitor.

(6) In some embodiments, controlling the first switching device to charge the first capacitor

includes controlling the first switching device to enable the energy storage device to provide current to the second inductor, wherein providing current to the second inductor includes inducing a voltage across the first inductor, and controlling the first switching device to disable the energy storage device from providing current to the second inductor, wherein the first inductor is configured to discharge to the first capacitor responsive to the switching device disabling the energy storage device.

(7) In at least one embodiment, controlling the second switching device to charge the second capacitor includes controlling the second switching device to enable the energy storage device to provide current to the fourth inductor, wherein providing current to the fourth inductor includes inducing a voltage across the third inductor, and controlling the second switching device to disable the energy storage device from providing current to the fourth inductor, wherein the third inductor is configured to discharge to the second capacitor responsive to the switching device disabling the energy storage device.

(8) In some embodiments, the output power includes an output waveform having a positive portion and a negative portion, and the positive portion of the output waveform is derived from power provided by the first capacitor, and the negative portion of the output waveform is derived from power provided by the second capacitor. In one embodiment, the primary branch portion is galvanically isolated from the backup branch portion. In some embodiments, the backup branch portion further includes a switching device having a first connection coupled to the energy storage device and a second connection switchably coupled to the second inductor, and a diode having an anode connection coupled to the first connection of the switching device and a cathode connection coupled to the second connection of the switching device.

(9) In at least one embodiment, the system further comprises a switching device including a first connection coupled to the output, and a second connection configured to be coupled to one of the output circuit or the first input, wherein the controller is configured to control the switching device to connect the second connection to the first input in a bypass mode of operation, and is configured to control the switching device to connect the second connection to the output circuit in the normal mode of operation and the backup mode of operation. In some embodiments, the system further comprises a first diode having a cathode connection coupled to the first inductor, and an anode connection coupled to a reference node, and a second diode having a cathode connection coupled to the reference node, and an anode connection coupled to the third inductor.

(10) According to at least one aspect of the disclosure, a system is provided including a first input configured to be coupled to an alternating current (AC) power source, a second input configured to be coupled to an energy storage device, an output configured to provide output power derived from at least one of the first input or the second input, an output circuit coupled to the output, the output circuit including a positive direct current (DC) bus and a negative DC bus, and means for independently providing power derived from at least one of the first input or the second input to the positive DC bus and the negative DC bus.

(11) In one embodiment, the output circuit includes a first capacitor and a second capacitor. In at least one embodiment, the system further includes means for charging the first capacitor at a first rate and means for charging the second capacitor at a second rate different than the first rate. In some embodiments, the system further includes means for charging the first capacitor and the second capacitor simultaneously. In an embodiment, the output power is AC power having a positive portion and a negative portion, and wherein the positive portion of the output power is derived from power provided by the first capacitor, and wherein the negative portion of the output power is derived from power provided by the second capacitor. In some embodiments, the second input is galvanically isolated from the output circuit. In at least one embodiment, the system further includes means for providing output power derived from the first input to the output and bypassing the output circuit.

(12) According to at least one aspect of the disclosure, a non-transitory computer-readable medium

storing thereon sequences of computer-executable instructions for controlling a power device is provided comprising a first input to receive first input power, a second input to receive second input power, an output to provide output power, an output circuit to provide the output power to the output, and a power conversion circuit including a primary branch portion having a first inductor, and a backup branch portion having a second inductor, the second inductor being magnetically coupled to the first inductor, the sequences of computer-executable instructions including instructions that instruct at least one processor to control one or more switching devices in the power device to receive, by the first inductor in a normal mode of operation, the first input power, provide, by the first inductor in the normal mode of operation, first power derived from the first input power to the output circuit, receive, by the second inductor in a backup mode of operation, the second input power, store, by the second inductor in the backup mode of operation, first stored energy derived from the second input power, store, by the first inductor in the backup mode of operation, second stored energy derived from the first stored energy, and provide, by the first inductor in the backup mode of operation, second power derived from the second stored energy to the output circuit.
(13) In some embodiments, the output circuit includes a first capacitor and a second capacitor, wherein the primary branch portion further includes a third inductor, and wherein the backup branch portion includes a fourth inductor magnetically coupled to the third inductor, the sequences of computer-executable instructions further including instructions that instruct the at least one processor to control the one or more switching devices to provide, by the first inductor and the third inductor, a first charging current to the first capacitor and a second charging current to the second capacitor.
(14) In at least one embodiment, the sequences of computer-executable instructions further including instructions that instruct the at least one processor to control the one or more switching devices to provide the first charging current to the first capacitor at a first rate and provide the second charging current to the second capacitor at a second rate, wherein the first rate is different than the second rate and wherein the first charging current is provided simultaneously with the second charging current.
(15) In some embodiments, the power device further includes a switching device having a first connection coupled to the output, and a second connection configured to be coupled to one of the output circuit or the first input, and wherein the sequences of computer-executable instructions further include instructions that instruct the at least one processor to control the switching device to connect the second connection to the first input in a bypass mode of operation, and to control the switching device to connect the second connection to the output circuit in the normal mode of operation and the backup mode of operation

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Various aspects of at least one embodiment are discussed below with reference to the accompanying figures, which are not intended to be drawn to scale. The figures are included to provide an illustration and a further understanding of the various aspects and embodiments, and are incorporated in and constitute a part of this specification, but are not intended as a definition of the limits of any particular embodiment. The drawings, together with the remainder of the specification, serve to explain principles and operations of the described and claimed aspects and embodiments. In the figures, each identical or nearly identical component that is illustrated in various figures is represented by a like numeral. For purposes of clarity, not every component may be labeled in every figure. In the figures:
(2) FIG. **1** illustrates a circuit diagram of a conventional online UPS;

(3) FIG. **2** illustrates a schematic diagram of a UPS according to an embodiment;
(4) FIG. **3** illustrates a process of controlling the UPS according to an embodiment;
(5) FIG. **4** illustrates a circuit diagram of the UPS including a first current path according to an embodiment;
(6) FIG. **5** illustrates a circuit diagram of the UPS including a second current path according to an embodiment;
(7) FIG. **6** illustrates a circuit diagram of the UPS including a third current path according to an embodiment;
(8) FIG. **7** illustrates a circuit diagram of the UPS including a fourth current path according to an embodiment;
(9) FIG. **8** illustrates a circuit diagram of the UPS including a fifth current path according to an embodiment;
(10) FIG. **9** illustrates a circuit diagram of the UPS including a sixth current path according to an embodiment;
(11) FIG. **10** illustrates a circuit diagram of the UPS including a seventh current path according to an embodiment;
(12) FIG. **11** illustrates a circuit diagram of the UPS including an eighth current path according to an embodiment;
(13) FIG. **12** illustrates a schematic diagram of a UPS according to an embodiment;
(14) FIG. **13** illustrates a schematic diagram of a UPS according to an embodiment; and
(15) FIG. **14** illustrates a schematic diagram of a UPS according to an embodiment.

DETAILED DESCRIPTION OF THE INVENTION

(16) Examples of the methods and systems discussed herein are not limited in application to the details of construction and the arrangement of components set forth in the following description or illustrated in the accompanying drawings. The methods and systems are capable of implementation in other embodiments and of being practiced or of being carried out in various ways. Examples of specific implementations are provided herein for illustrative purposes only and are not intended to be limiting. In particular, acts, components, elements and features discussed in connection with any one or more examples are not intended to be excluded from a similar role in any other examples.

(17) Also, the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. Any references to examples, embodiments, components, elements or acts of the systems and methods herein referred to in the singular may also embrace embodiments including a plurality, and any references in plural to any embodiment, component, element or act herein may also embrace embodiments including only a singularity. References in the singular or plural form are no intended to limit the presently disclosed systems or methods, their components, acts, or elements. The use herein of “including,” “comprising,” “having,” “containing,” “involving,” and variations thereof is meant to encompass the items listed thereafter and equivalents thereof as well as additional items. References to “or” may be construed as inclusive so that any terms described using “or” may indicate any of a single, more than one, and all of the described terms. In addition, in the event of inconsistent usages of terms between this document and documents incorporated herein by reference, the term usage in the incorporated features is supplementary to that of this document; for irreconcilable differences, the term usage in this document controls.

(18) As discussed above, certain UPSs, including online UPSs, can include multiple power conversation stages. For example, a conventional online UPS may include an input PFC converter and a separate DC/DC converter coupled to an energy storage device to convert the input AC power and stored DC power, respectively. FIG. **1** illustrates a circuit diagram of a conventional online UPS **100** including multiple input power conversion stages. The online UPS **100** includes a PFC converter **102**, a DC/DC converter **104**, an inverter **106**, and a controller **107**. An input of the PFC converter **102** is configured to be coupled to an AC power source **108** and an output of the

PFC converter **102** is coupled to the inverter **106** via a first DC bus **110** and a second DC bus **112**. The DC/DC converter **104** is configured to be coupled to an energy storage device **114** and is coupled to the inverter **106** via the first DC bus **110** and the second DC bus **112**. The inverter **106** is configured to be coupled to, and provide output power to, a load **116**. The controller **107** is communicatively coupled to, and may provide control signals to, one or more switching devices (for example, transistors) in the UPS **100**.

(19) Responsive to receiving input AC power from the AC power source **106**, the PFC converter **102** is configured to convert the received AC power to DC power and provide the DC power to the inverter **106** via the first DC bus **110** and the second DC bus **112**. If the energy storage device **114** is not fully charged, then the PFC converter **102** may also provide a portion of the DC power to the DC/DC converter **104** to charge the energy storage device **114** via the first DC bus **110** and the second DC bus **112**.

(20) Responsive to the AC power source **106** being unable to provide sufficient AC power to the PFC converter **102** (for example, due to a blackout or brownout condition), the DC/DC converter **104** may be configured to enter a backup mode of operation. In the backup mode of operation, the DC/DC converter **104** may be configured to draw backup DC power from the energy storage device **114**, convert the backup DC power to converted DC power (for example, by changing a voltage level of the energy stored in the energy storage device **114**), and provide the converted DC power to the inverter **106** via the first DC bus **110** and the second DC bus **112**. The inverter **106** is configured to convert DC power from the first DC bus **110** and the second DC bus **112** (derived from at least one of the input AC power or the backup DC power as described above) into output AC power and the output AC power is provided to the load **116**.

(21) A size, component count, and cost of the conventional online UPS **100** may be disadvantageously high, for example, in part due to the PFC converter **102** and the DC/DC converter **104** being implemented as separate converters. The PFC converter **102** may be inactive when the DC/DC converter **104** is active, and the DC/DC converter **104** may be inactive when the PFC converter **102** is active. Thus, it may be beneficial to provide a single converter topology capable of performing functions of the PFC converter **102** and the DC/DC converter **104** without extended periods of inactivity to reduce the size, component count, and cost of an online UPS in which such a converter would be implemented.

(22) The conventional online UPS **100** may also be disadvantageous because the power provided to the first DC bus **110** and the second DC bus **112** typically is not independently controlled in the backup mode of operation. For example, power provided by the PFC converter **102** and the DC/DC converter **104** to the first DC bus **110** and the second DC bus **112** is provided in equal proportions, without regard for the demands of the load **116**. This may result in inefficient operation of the conventional online UPS **100**. For instance, the conventional online UPS **100** may be unable to efficiently provide power to a load requiring an uneven distribution of power from the busses **110**, **112** in a backup mode of operation, such as a load requiring a half-wave rectified waveform drawn primarily from the first DC bus **110**.

(23) In view of the foregoing, the conventional online UPS **100** may be disadvantageously costly, large, and inflexible. Accordingly, aspects of the present disclosure provide a topology capable of addressing at least some of the foregoing deficiencies to reduce costs and improve efficiency of a UPS.

(24) FIG. **2** illustrates a schematic diagram of a UPS **200** according to one embodiment described herein. The UPS **200** includes a dual converter **202**, an inverter **204**, and a controller **205**. The dual converter **202** is configured to be coupled to an AC power source **206**, and is coupled to the inverter **204**. The inverter **204** is coupled to the dual converter **202**, and may be configured to be coupled to a load.

(25) The dual converter **202** includes a primary branch **208** and a backup branch **210**. The primary branch **208** includes a first relay **212**, a first diode **214**, a second diode **216**, a first inductor **218**, a

second inductor **220**, a first switching device **222**, a third diode **224**, and a fourth diode **226**. The backup branch **210** includes an energy storage device **228**, a second relay **230**, a third inductor **232**, a fourth inductor **234**, a second switching device **236**, and a third switching device **238**. In some examples, the energy storage device **228** may be external to the dual converter **202** and may be coupled to the backup branch **210**. The inverter **204** may include a fifth diode **240**, a sixth diode **242**, a first capacitor **244**, a second capacitor **246**, a fourth switching device **248**, a fifth switching device **250**, a fifth inductor **252**, and a third capacitor **254**.

(26) In some embodiments, other inverter topologies may be implemented in lieu of, or in addition to, the topology illustrated in connection with the inverter **204**. In other embodiments, the inverter **204** may be omitted such that the UPS **200** provides DC, rather than AC, output power. In still other embodiments, the components **240-254** of the inverter **204** may be implemented as illustrated in FIG. **2**, but the switching devices **248**, **250** be operated to provide DC, rather than AC, output power. For example, only one of the switching devices **248**, **250** may be closed and conducting while providing output power to the load **256**, such that power having only one voltage polarity is provided to the load **256**. Thus, although the components **240-254** are described in one example as being components of the inverter **204**, the components **240-254** may be operated to provide either DC or AC output power as required or desired by the load **256**. The inverter **204**, or a circuit implemented in lieu of the inverter **204**, may be referred to herein as an “output circuit.”

(27) In at least one embodiment, the first relay **212** is configured as a single-pole double-throw switching device having a first terminal, a second terminal, and a third terminal. The third terminal is configured to be switchably connected to one of the first terminal and the second terminal. The first terminal is configured to be coupled to the AC power source **206**. The second terminal is coupled to a reference node **213** (for example, a node at a reference voltage such as a ground voltage). The third terminal is configured to be coupled to the first diode **214** and the second diode **216**. The first diode **214** has an anode coupled to the first relay **212** and the second diode **216**, and a cathode coupled to the first inductor **218**. The second diode **216** has a cathode coupled to the first relay **212** and the first diode **214**, and an anode coupled to the second inductor **220**.

(28) The first inductor **218** is coupled to the first diode **214** at a first connection, and is coupled to the first switching device **222**, the third diode **224**, and the fifth diode **240** at a second connection. The first inductor **218** is further configured to be magnetically coupled to the third inductor **232**. As used herein, “magnetically coupled” may refer to a relationship of at least two inductive components in which a magnetic field and/or a change in a magnetic field produced by a first inductive component induces a voltage across a second inductive component (for example, mutual inductance).

(29) The second inductor **220** is coupled to the second diode **216** at a first connection, and is coupled to the first switching device **222**, the fourth diode **226**, and the sixth diode **242** at a second connection. The second inductor **220** is further configured to be magnetically coupled to the fourth inductor **234**. The first switching device **222** is coupled to the first inductor **218**, the third diode **224**, and the fifth diode **240** at a first connection, is coupled to the second inductor **220**, the fourth diode **226**, and the sixth diode **242** at a second connection, and is configured to be communicatively coupled to the controller **205** at a control connection. The third diode **224** includes a cathode connection coupled to the first inductor **218**, the first switching device **222**, and the fifth diode **240**, and an anode connection coupled to the fourth diode **226** and the reference node **213**. The fourth diode **226** includes a cathode connection coupled to the third diode **224** and the reference node **213**, and an anode connection coupled to the second inductor **220**, the first switching device **222**, and the sixth diode **242**.

(30) The energy storage device **228** is coupled to the second relay **230** at a first connection, and is coupled to the second switching device **236** and the third switching device **238** at a second connection. In at least one embodiment, the second relay **230** is configured as a single-pole single-throw switch having a first terminal, a second terminal, and a third terminal. The third terminal is

configured to be switchably connected to one of the first terminal and the second terminal. The first terminal is configured to be coupled to the third inductor **232** and the fourth inductor **234**. In an embodiment, the second terminal may not be permanently connected to any other component, and is configured to be switchably connected to the third terminal. Accordingly, in an embodiment, current does not pass through the second terminal. The third terminal is configured to be coupled to the energy storage device **228**. In some examples, the energy storage device **228** may be coupled to a charger (not illustrated) configured to charge the energy storage device **228**. For example, a charger may be coupled in parallel with the energy storage device **228**.
(31) The third inductor **232** is coupled to the second relay **230** at a first connection, and is coupled to the second switching device **236** at a second connection. The fourth inductor **234** is coupled to the second relay **230** at a first connection, and is coupled to the third switching device **238** at a second connection. The second switching device **236** is coupled to the third inductor **232** at a first connection, is coupled to the energy storage device **228** at a second connection, and is configured to be communicatively coupled to the controller **205** at a control connection. The third switching device **238** is coupled to the fourth inductor **234** at a first connection, is coupled to the energy storage device **228** at a second connection, and is configured to be communicatively coupled to the controller **205** at a control connection.
(32) The fifth diode **240** has an anode coupled to the first inductor **218**, the first switching device **222**, and the third diode **224**, and a cathode coupled to the first capacitor **244** and the fourth switching device **248**. The sixth diode **242** has a cathode coupled to the second inductor **220**, the first switching device **222**, and the fourth diode **226**, and an anode coupled to the second capacitor **246** and the fifth switching device **250**. The first capacitor **244** is coupled to the fifth diode **240** and the fourth switching device **248** at a first connection, and is coupled to the reference node **213** at a second connection. The second capacitor **246** is coupled to the reference node **213** at a first connection, and is coupled to the sixth diode **242** and the fifth switching device **250** at a second connection.
(33) The fourth switching device **248** is coupled to the fifth diode **240** and the first capacitor **244** at a first connection, is coupled to the fifth switching device **250** and the fifth inductor **252** at a second connection, and is configured to be communicatively coupled to the controller **205** at a control connection. The fifth switching device **250** is coupled to the fourth switching device **248** and the fifth inductor **252** at a first connection, is coupled to the sixth diode **242** and the second capacitor **246** at a second connection, and is configured to be communicatively coupled to the controller **205** at a control connection.
(34) The fifth inductor **252** is coupled to the fourth switching device **248** and the fifth switching device **250** at a first connection, and is configured to be coupled to the third capacitor **254** at a second connection. The third capacitor **254** is coupled to the fifth inductor **252** at a first connection, and is coupled to the reference node **213** at a second connection. In some examples, the third capacitor **254** may be configured to be coupled to a load external to the UPS **200**. For example, the third capacitor **254** may be configured to be coupled in parallel with a load.
(35) The dual converter **202** is configured to receive input power from at least one of the AC power source **206** and the energy storage device **228**, convert the received power to DC power (for example, from AC power or DC power of a different voltage level), and provide the converted DC power to the inverter **204**. The inverter **204** is configured to receive the DC power from the dual converter **202**, convert the received DC power to output AC power, and provide the output AC power to a load. The controller **205** is configured to control the dual converter **202** and the inverter **204**. For example, controlling operation of the dual converter **202** and the inverter **204** may include controlling a switching operation of the first switching device **222**, the second switching device **236**, the third switching device **238**, the fourth switching device **248**, and the fifth switching device **250**.
(36) The controller **205** may determine a mode of operation of the UPS **200** and, based on the mode

of operation, control the dual converter **202** to draw power from at least one of the AC power source **206** or the energy storage device **228**. For example, the controller **205** may determine a quality of power provided by the AC power source **206** based on one or more received parameters (for example, voltage, current, frequency, etc.) indicative of the quality of power provided by the AC power source **206**. Determining a quality of power provided by the AC power source **206** may include, for example, determining if a parameter indicative of the power quality (for example, a voltage or current parameter) is within a permissible range.

(37) If the parameter indicative of the power quality is within the permissible range, the controller **205** may determine that the UPS **200** is in a normal mode of operation characterized at least partially by the UPS **200** receiving acceptable-quality power from the AC power source **206**. In the normal mode of operation, the controller **205** may control the dual converter **202** to draw power from the AC power source **206**. If the parameter indicative of the power quality is outside of the permissible range, the controller **205** may determine that the UPS **200** is in a backup mode of operation characterized at least partially by the UPS **200** being unable to receive acceptable-quality power from the AC power source **206**. In the backup mode of operation, the controller **205** may control the dual converter **202** to draw power from the energy storage device **228**.

(38) FIG. **3** illustrates a process **300** of controlling the UPS **200** according to an embodiment. At act **302**, the process **300** begins. At act **304**, a determination is made as to whether the UPS **200** is in a normal mode of operation. For example, as discussed above, the controller **205** may determine whether the UPS **200** is in the normal mode of operation based on one or more parameters indicative of a quality of power provided by the AC power source **206**. If the controller **205** determines that the UPS **200** is in a normal mode of operation (**304** YES), then the process **300** continues to act **306**.

(39) At act **306**, the controller **205** controls the first relay **212** and the second relay **230**. Controlling the first relay **212** includes connecting the third terminal of the first relay **212** to the first terminal of the first relay **212** such that the AC power source **206** is coupled to the first diode **214** and the second diode **216** via the first relay **212**. Controlling the second relay **230** includes connecting the third terminal of the second relay **230** to the second terminal of the second relay **230** such that the energy storage device **228** is disconnected from the third inductor **232** and the fourth inductor **234**.

(40) At act **308**, the first inductor **218** is energized. In one embodiment, the first inductor **218** is energized at act **308** by AC power received from the AC power source **206** where the AC power is in a positive half-cycle of a sinusoidal waveform. In one example, the first diode **214** is forward-biased and the second diode **216** is reverse-biased during a positive half-cycle of power provided by the AC power source **206**. Energizing the first inductor **218** includes controlling, by the controller **205**, the first switching device **222** to alternate between an open and non-conducting position and a closed and conducting position. The controller **205** may control the first switching device **222** in accordance with the power received from the AC power source **206**. For example, the controller **205** may control the first switching device **222** to maintain a sinusoidal current through the first switching device **222** from the AC power source **206**.

(41) When the first switching device **222** is in a closed and conducting position, AC power provided by the AC power source **206** is delivered to a conductive path including the AC power source **206**, the first relay **212**, the first diode **214**, the first inductor **218**, the first switching device **222**, and the fourth diode **226**. FIG. **4** illustrates a circuit diagram of the UPS **200** indicating a current path **400** at act **308**.

(42) At act **310**, the first inductor **218** is de-energized. For example, act **310** includes controlling, by the controller **205**, the first switching device **222** to transition from a closed and conducting position to an open and non-conducting position, thereby interrupting the current path **400**. The first inductor **218** discharges to a conductive path including the AC power source **206**, the first relay **212**, the first diode **214**, the first inductor **218**, the fifth diode **240**, and the first capacitor **244** to charge the first capacitor **244**. FIG. **5** illustrates a circuit diagram of the UPS **200** indicating a

current path **500** at act **310**.

(43) As discussed above, the first inductor **218** is magnetically coupled to the third inductor **232**. However, during acts **308** and **310**, the second switching device **236** is maintained in an open and non-conducting position. Accordingly, although a voltage is induced across the third inductor **232**, no induced current passes through the third inductor **232** during acts **308** and **310** at least because the third inductor **232** is coupled in series with the open and non-conducting second switching device **236**.

(44) At act **312**, the second inductor **220** is energized. In one embodiment, the second inductor **220** is energized at act **312** by power received from the AC power source **206** where the AC power provided by the AC power source **206** is in a negative half-cycle of the sinusoidal waveform. In one example, the first diode **214** is reverse-biased and the second diode **216** is forward-biased during a negative half-cycle of power provided by the AC power source **206**. Energizing the second inductor **220** may include controlling, by the controller **205**, the first switching device **222** to be in a closed and conducting position such that power provided by the AC power source **206** is delivered to a conductive path including the AC power source **206**, the second diode **216**, the second inductor **220**, the first switching device **222**, and the third diode **224**. Controlling the first switching device **222** at act **312** may be executed similarly to controlling the first switching device **222** as discussed above with respect to act **308**. FIG. **6** illustrates a circuit diagram of the UPS **200** indicating a current path **600** at act **312**.

(45) At act **314**, the second inductor **220** is de-energized. For example, act **314** includes controlling, by the controller **205**, the first switching device **222** to transition from a closed and conducting position to an open and non-conducting position, thereby interrupting the current path **600**. The second inductor **220** discharges to a conductive path including the AC power source **206**, the first relay **212**, the second diode **216**, the second inductor **220**, the sixth diode **242**, and the second capacitor **246** to charge the second capacitor **246**. FIG. **7** illustrates a circuit diagram of the UPS **200** indicating a current path **700** at act **314**.

(46) As discussed above, the second inductor **220** is magnetically coupled to the fourth inductor **234**. However, during acts **312** and **314**, the third switching device **238** is maintained in an open and non-conducting position. Accordingly, although a voltage is induced across the fourth inductor **234**, no induced current passes through the fourth inductor **234** during acts **312** and **314** at least because the fourth inductor **234** is coupled in series with the open and non-conducting third switching device **238**.

(47) Accordingly, acts **308** and **310** are executed during a positive half-cycle of input power received from the AC power source **206** to charge the first capacitor **244**, and acts **312** and **314** are executed during a negative half-cycle of input power received from the AC power source **206** to charge the second capacitor **246**. The inverter **204** may be controlled during execution of acts **308-314** to draw power from the first capacitor **244** and the second capacitor **246**, convert the power to AC power, and provide the AC power to an output. For example, the controller **205** may use Pulse Width Modulation (PWM) in connection with control signals provided to the switches **248**, **250** to provide the AC power to the output. After act **314**, the process **300** returns to act **304**.

(48) Responsive to determining that the UPS **200** is still in the normal mode of operation (**304** YES), acts **306-314** are repeated. Responsive to determining that the UPS **200** is not in the normal mode of operation (**304** NO), the process **300** continues to act **316**. At act **316**, the controller **205** controls the first relay **212** and the second relay **230**. Controlling the first relay **212** may include connecting the third terminal of the first relay **212** to the second terminal of the first relay **212** such that the AC power source **206** is disconnected from the UPS **200**, and the first diode **214** and the second diode **216** are coupled to the reference node **213**. Controlling the second relay **230** may include connecting the third terminal of the second relay **230** to the first terminal of the second relay **230** such that the energy storage device **228** is connected to the third inductor **232** and the fourth inductor **234**.

(49) At act **318**, the first inductor is energized. For example, act **318** may include controlling, by the controller **205**, the second switching device **236** to be in a closed and conducting position to energize the third inductor **232**. Because the third inductor **232** is magnetically coupled to the first inductor **218**, a voltage is induced across the first inductor **218**, thereby energizing the magnetically coupled first inductor **218**. Energizing the third inductor **232**, and consequently the first inductor **218**, may include providing, by the energy storage device **228**, power to the third inductor **232** in a conductive path including the energy storage device **228**, the second relay **230**, the third inductor **232**, and the second switching device **236**. FIG. **8** illustrates a circuit diagram of the UPS **200** indicating a current path **800** at act **318**.
(50) Although the first inductor **218** is energized by the energization of the magnetically coupled third inductor **232**, the first diode **214** prevents the first inductor **218** from discharging. The induced voltage across the first inductor **218** reverse-biases the first diode **214**, which is coupled in series with the first inductor **218**, thereby preventing any induced current from passing through the first inductor **218** during act **318**.
(51) At act **320**, the first inductor is de-energized. For example, act **320** may include controlling, by the controller **205**, the second switching device **236** to be in an open and non-conducting position, thereby preventing the energy storage device **228** from discharging to the third inductor **232**. Responsive to the cessation of current through the third inductor **232**, the polarity of the induced voltage across the first inductor **218** is reversed, thereby forward-biasing the first diode **214**.
(52) The first inductor **218** is therefore able to discharge current to a conductive path including the first relay **212**, the first diode **214**, the first inductor **218**, the fifth diode **240**, and the first capacitor **244** to charge the first capacitor **244**. FIG. **9** illustrates a circuit diagram of the UPS **200** indicating a current path **900** at act **320**.
(53) At act **322**, the second inductor is energized. For example, act **322** may include controlling, by the controller **205**, the third switching device **238** to be in a closed and conducting position to energize the fourth inductor **234**. Because the fourth inductor **234** is magnetically coupled to the second inductor **220**, a voltage is induced across the second inductor **220**, thereby energizing the magnetically coupled second inductor **220**.
(54) Energizing the fourth inductor **234**, and consequently the second inductor **220**, may include providing, by the energy storage device **228**, power to the fourth inductor **234** in a conductive path including the energy storage device **228**, the second relay **230**, the fourth inductor **234**, and the third switching device **238**. FIG. **10** illustrates a circuit diagram of the UPS **200** indicating a current path **1000** at act **322**.
(55) Although the second inductor **220** is energized by the energization of the magnetically coupled fourth inductor **234**, the second diode **216** prevents the second inductor **220** from discharging. The induced voltage across the second inductor **220** reverse-biases the second diode **216**, which is coupled in series with the second inductor **220**, thereby preventing any induced current from passing through the second inductor **220** during act **322**.
(56) At act **324**, the second inductor is de-energized. For example, act **324** may include controlling, by the controller **205**, the third switching device **238** to be in an open and non-conducting position, thereby preventing the energy storage device **228** from discharging to the fourth inductor **234**. Responsive to the cessation of current through the fourth inductor **234**, the polarity of the induced voltage across the second inductor **220** is reversed, thereby forward-biasing the second diode **216**.
(57) The second inductor **220** is therefore able to discharge current to a conductive path including the first relay **212**, the second diode **216**, the second inductor **220**, the sixth diode **242**, and the second capacitor **246** to charge the second capacitor **246**. FIG. **11** illustrates a circuit diagram of the UPS **200** indicating a current path **1100** at act **324**.
(58) Accordingly, acts **318** and **320** are executed to provide input power from the energy storage device **228** to charge the first capacitor **244**, and acts **322** and **324** are executed to provide input power from the energy storage device **228** to charge the second capacitor **246**. The inverter **204**

may be controlled during execution of acts **318**-**324** to draw power from the first capacitor **244** and the second capacitor **246**, convert the power to AC power, and provide the AC power to an output. For example, the controller **205** may use Pulse Width Modulation (PWM) in connection with control signals provided to the switches **248**, **250** to provide the AC power to the output. After act **324**, the process **300** returns to act **304**.

(59) Accordingly, process **300** may be executed by the UPS **200** to provide output power to a load during a normal mode of operation (for example, when the quality of AC power received from the AC power source **206** is acceptable) and during a backup mode of operation (for example, when the quality of AC power received from the AC power source **206** is not acceptable). As discussed above, the UPS **200** may be advantageous relative to certain UPSs, such as the conventional online UPS **100**, at least because a component count of the UPS **200** is reduced relative to the conventional online UPS **100**. The cost and physical footprint of the UPS **200** may therefore be reduced relative to the conventional online UPS **100**.

(60) In at least one embodiment, the primary branch **208** may be galvanically isolated from the backup branch **210**. As used herein, "galvanically isolated" may refer to a relationship between at least two components in which no direct current path exists between the at least two components. For example, the primary branch **208** may be magnetically coupled to the backup branch **210** via the first inductor **218**, the second inductor **220**, the third inductor **232**, and the fourth inductor **234**, but does not have a direct current path via a physical conductive medium, such as a wire, bus, or other solid conductor. Moreover, the primary branch **208** and the backup branch **210** may not share a common ground or return connection. Accordingly, the primary branch **208** may be referred to herein as being galvanically isolated from the backup branch **210**.

(61) Moreover, a respective amount of power provided to the first capacitor **244** and the second capacitor **246** may be individually controlled. For example, during the normal mode of operation, the amount of power provided to the first capacitor **244** at act **310** (for example, the amount of power provided during a positive half-cycle of the input AC power) may be proportional to an amount of energy provided to the first inductor **218** at act **308**. The amount of energy provided to the first inductor **218** may in turn be proportional to a duration of controlling the first switching device **222** to be in a closed and conducting position (for example, by controlling a duty cycle of a control signal provided to the first switching device **222** by the controller **205**). Accordingly, an amount of power provided to the first capacitor **244** at act **310** may be individually controlled by the controller **205** at act **308**.

(62) Similar principles apply to the amount of power provided to the second capacitor **246** at acts **312** and **314**. For example, during the normal mode of operation, the amount of power provided to the second capacitor **246** at act **314** (for example, the amount of power provided during a negative half-cycle of the input AC power) may be proportional to an amount of energy provided to the second inductor **220** at act **312**. The amount of energy provided to the second inductor **220** may in turn be proportional to a duration of controlling the first switching device **222** to be in a closed and conducting position (for example, by controlling the duty cycle of the control signal provided to the first switching device **222** by the controller **205**). Accordingly, an amount of power provided to the second capacitor **246** at act **314** may be controlled by the controller **205** at act **312**.

(63) In some embodiments, the first capacitor **244** and the second capacitor **246** may be individually charged and discharged. As used herein, "individual charging and discharging" may refer to charging or discharging one component without directly affecting, or being constrained by, another component. Accordingly, the first capacitor **244** may be charged or discharged at a first rate, and the second capacitor **246** may be charged or discharged at a second rate, where the first rate is independent of the second rate.

(64) During the backup mode of operation, the amount of power provided to the first capacitor **244** at act **318** (for example, the amount of power provided by the energy storage device **228** to the first capacitor **244**) may be proportional to an amount of energy provided to the first inductor **218** at act

**318**. The amount of energy provided to the first inductor **218** may in turn be proportional to a duration of controlling the second switching device **236** to be in a closed and conducting position (for example, by controlling a duty cycle of a control signal provided to the second switching device **236** by the controller **205**), thereby energizing the third inductor **232**. Accordingly, an amount of power provided to the first capacitor **244** at act **320** may be controlled by the controller **205** at act **318**.

(65) Similar principles apply to the amount of power provided to the second capacitor **246** at acts **322** and **324**. For example, during the backup mode of operation, the amount of power provided to the second capacitor **246** at act **324** (for example, the amount of power provided by the energy storage device **228** to the second capacitor **246**) may be proportional to an amount of energy provided to the second inductor **220** at act **322**. The amount of energy provided to the second inductor **220** may in turn be proportional to a duration of controlling the third switching device **238** to be in a closed and conducting position (for example, by controlling a duty cycle of a control signal provided to the third switching device **238** by the controller **205**). Accordingly, an amount of power provided to the second capacitor **246** at act **324** may be controlled by the controller **205** at act **322**.

(66) An amount of power provided to the first capacitor **244** and an amount of power provided to the second capacitor **246** may be individually controlled. Individual control of the amount of power provided to the first capacitor **244** and the second capacitor **246** may enable the UPS **200** to respond to load requirements more effectively. For example, if the UPS **200** provides power to a load requiring rectified power with a positive voltage, the UPS **200** may provide additional power to the first capacitor **244** to compensate for an increased amount of power being drawn from the first capacitor **244** by the inverter **206** to generate the rectified power.

(67) FIG. **12** illustrates a schematic diagram of a UPS **1200** according to an embodiment. The UPS **1200** includes a dual converter **1202**, an inverter **1204**, and a controller **1205**. The dual converter **1202** is configured to be coupled to an AC input **1206**, and is coupled to the inverter **1204**. The inverter **1204** is coupled to the dual converter **1202**, and may be configured to be coupled to a load. The dual converter **1202** includes a primary branch **1208** and a backup branch **1210**.

(68) The UPS **1200** is similar to the UPS **200**. For example, the inverter **1204**, the controller **1205**, and the primary branch **1208** are similar to the inverter **204**, the controller **205**, and the primary branch **208**. The backup branch **1210** is similar to the backup branch **210** with the second switching device **236** and the third switching device **238** of the backup branch **210** replaced by a single switching device in the backup branch **1210**.

(69) More specifically, the backup branch **1210** includes an energy storage device **1212**, a relay **1214**, a first inductor **1216**, a second inductor **1218**, and a switching device **1220**. The energy storage device **1212** is coupled to the relay **1214** at a first connection, and is coupled to the switching device **1220** at a second connection. The relay **1214** is configured as a single-pole single-throw switch having a first terminal, a second terminal, and a third terminal. The third terminal is configured to be switchably connected to one of the first terminal and the second terminal. The first terminal is configured to be coupled to the first inductor **1216** and the second inductor **1218**. In an embodiment, the second terminal may not be permanently connected to any other component, and is configured to be switchably connected to the third terminal. Accordingly, in an embodiment, current does not pass through the second terminal. The third terminal is configured to be coupled to the energy storage device **1212**.

(70) The first inductor **1216** is coupled to the relay **1214** at a first connection, and is coupled to the switching device **1220** at a second connection. The second inductor **1218** is coupled to the relay **1214** at a first connection, and is coupled to the switching device **1220** at a second connection. The switching device **1220** is coupled to the first inductor **1216** and the second inductor **1218** at a first connection, is coupled to the energy storage device **1212** at a second connection, and is configured to be communicatively coupled to the controller **1205** at a control connection.

(71) In some examples, the UPS **200** and the UPS **1200** are configured to operate similarly. For example, the UPSs **200**, **1200** may operate in a substantially similar fashion during a normal mode of operation. During a backup mode of operation, the first inductor **1216** and the second inductor **1218** are not independently energized because the first inductor **1216** and the second inductor **1218** are connected to the single, shared switching device **1220**. In other words, the first inductor **1216** is not energized without energizing the second inductor **1218**, and the second inductor **1218** is not energized without energizing the first inductor **1216**.
(72) In contrast, the third inductor **232** can be energized independently of the fourth inductor **234** by closing the second switching device **236** and opening the third switching device **238**, and the fourth inductor **234** can be energized independently of the third inductor **232** by closing the third switching device **238** and opening the second switching device **236**. However, a component count of the UPS **1200** is less than a component count of the UPS **200** at least because the number of switching devices is reduced by one, thereby reducing the cost and footprint of the UPS **1200** relative to the UPS **200**.
(73) FIG. **13** illustrates a schematic diagram of a UPS **1300** according to an embodiment. The UPS **1300** includes a dual converter **1302**, an inverter **1304**, and a controller **1305**. The dual converter **1302** is configured to be coupled to an AC input **1306**, and is coupled to the inverter **1304**. The inverter **1304** is coupled to the dual converter **1302**, and may be configured to be coupled to a load. The dual converter **1302** includes a primary branch **1308** and a backup branch **1310**.
(74) The UPS **1300** is similar to the UPS **200**. For example, the inverter **1304** and the controller **1305** are similar to the inverter **204** and the controller **205**. The primary branch **1308** is similar to the primary branch **208** and has two additional diodes, and the backup branch **1310** is similar to the backup branch **210** and has an additional diode. More specifically, the primary branch **1308** includes a first relay **1312**, a first diode **1314**, a second diode **1316**, a third diode **1318**, a fourth diode **1320**, a first inductor **1322**, a second inductor **1324**, a first switching device **1326**, a fifth diode **1328**, and a sixth diode **1330**. The backup branch **1310** includes an energy storage device **1332**, a second relay **1334**, a seventh diode **1336**, a third inductor **1338**, a fourth inductor **1340**, a second switching device **1342**, and a third switching device **1344**. In another embodiment, the primary branch **1308** may be replaced by an alternate topology, such as that of the primary branch **208**. In yet another embodiment, the backup branch **1310** may be replaced by an alternate topology, such as that of the backup branch **210**.
(75) The first relay **1312** is configured as a single-pole double-throw switching device having a first terminal, a second terminal, and a third terminal. The third terminal is configured to be switchably connected to one of the first terminal and the second terminal. The first terminal is configured to be coupled to the AC input **1306**. The second terminal is coupled to a reference node **1313** (for example, a node at a reference voltage such as a ground voltage). The third terminal is configured to be coupled to the first diode **1314** and the second diode **1316**. The first diode **1314** has an anode coupled to the first relay **1312** and the second diode **1316**, and a cathode coupled to the first inductor **1322** and the third diode **1318**. The second diode **1316** has a cathode coupled to the first relay **1312** and the first diode **1314**, and an anode coupled to the fourth diode **1320** and the second inductor **1324**.
(76) The third diode **1318** has a cathode coupled to the first diode **1314** and the first inductor **1322**, and an anode coupled to the fourth diode **1320** and the reference node **1313**. The fourth diode **1320** has a cathode coupled to the third diode **1318** and the reference node **1313**, and an anode coupled to the second diode **1316** and the second inductor **1324**. The first inductor **1322** is coupled to the first diode **1314** and the third diode **1318** at a first connection, and is coupled to the first switching device **1326**, the fifth diode **1328**, and an eighth diode **1346** of the inverter **1304** at a second connection. The first inductor **1322** is further configured to be magnetically coupled to the third inductor **1338** of the backup branch **1310**.
(77) The second inductor **1324** is coupled to the second diode **1316** and the fourth diode **1320** at a

first connection, and is coupled to the first switching device **1326**, the sixth diode **1330**, and a ninth diode **1348** of the inverter **1304** at a second connection. The second inductor **1324** is further configured to be magnetically coupled to the fourth inductor **1340** of the backup branch **1310**. The first switching device **1326** is coupled to the first inductor **1322**, the fifth diode **1328**, and the eighth diode **1346** at a first connection, is coupled to the second inductor **1324**, the sixth diode **1330**, and the ninth diode **1348** at a second connection, and is configured to be communicatively coupled to the controller **1305** at a control connection.

(78) The fifth diode **1328** includes a cathode connection coupled to the first inductor **1322**, the first switching device **1326**, and the eighth diode **1346**, and an anode connection coupled to the sixth diode **1330** and the reference node **1313**. The sixth diode **1330** includes a cathode connection coupled to the fifth diode **1328** and the reference node **1313**, and an anode connection coupled to the second inductor **1324**, the first switching device **1326**, and the ninth diode **1348**.

(79) The second relay **1334** is configured as a single-pole single-throw switching device having a first terminal, a second terminal, and a third terminal. The third terminal is configured to be switchably connected to one of the first terminal and the second terminal. The first terminal is configured to be coupled to the seventh diode **1336**, the third inductor **1338**, and the fourth inductor **1340**. In an embodiment, the second terminal may not be permanently connected to any other component, and is configured to be switchably connected to the third terminal. Accordingly, in an embodiment, current does not pass through the second terminal. The third terminal is configured to be coupled to the energy storage device **1332** and to the seventh diode **1336**.

(80) The seventh diode **1336** includes an anode connection configured to be coupled to the third terminal of the second relay **1334**, and a cathode connection configured to be coupled to the first terminal of the second relay **1334**. The third inductor **1338** is coupled to the first terminal of the second relay **1334** at a first connection, and is coupled to the second switching device **1342** at a second connection. The third inductor **1338** is further configured to be magnetically coupled to the first inductor **1322**. The second inductor **1340** is coupled to the first terminal of the second relay **1334** at a first connection, and is coupled to the third switching device **1344** at a second connection. The fourth inductor **1340** is further configured to be magnetically coupled to the second inductor **1324**.

(81) The second switching device **1342** is configured to be coupled to the third inductor **1338** at a first connection, is configured to be coupled to the energy storage device **1332** at a second connection, and is configured to be communicatively coupled to the controller **1305** at a control connection. The third switching device **1344** is configured to be coupled to the fourth inductor **1340** at a first connection, is configured to be coupled to the energy storage device **1332** at a second connection, and is configured to be communicatively coupled to the controller **1305** at a control connection. The energy storage device **1332** is coupled to the third terminal of the second relay **1334** at a first connection, and is coupled to the third inductor **1342** and the fourth inductor **1344** at a second connection.

(82) As discussed above, the primary branch **1308** is similar to the backup branch **208** and includes the third diode **1318** and the fourth diode **1320**. The third diode **1318** and the fourth diode **1320** provide power walk-in. Power walk-in may be advantageous where the UPS **1300** transitions from a backup mode of operation to a normal mode of operation. Subsequent to the transition from the backup mode of operation to the normal mode of operation, the UPS **1300** may resume drawing power from the AC input **1306**. The UPS **1300** may draw a significant amount of current from the AC input **1306** upon being connected to the AC input **1306**. Power walk-in refers to a feature of the UPS **1306** by which the current drawn from the AC input **1306** is more gradually increased when transitioning from the backup mode of operation to the normal mode of operation.

(83) As discussed above, the third diode **1318** and the fourth diode **1320** are connected to the reference node **1313**. The third diode **1318** and the fourth diode **1320** enable power walk-in by coupling components of the UPS **1300** (including, for example, at least one of the first diode **1314**,

the second diode **1316**, the first inductor **1322**, and the second inductor **1324**) to the reference node **1313** via the third diode **1318** and the fourth diode **1320** while the UPS **1300** transitions from the backup mode of operation to the normal mode of operation. Because the components are coupled to the reference node **1313** during the transition, an amount of current drawn by the UPS **1300** from the AC source **1306** upon completion of the transition from the backup mode of operation to the normal mode of operation is reduced.

(84) As discussed above, the backup branch **1310** is similar to the backup branch **210** and additionally includes the seventh diode **1336**, having a cathode connection coupled to the first terminal of the second relay **1334**, which is coupled to the third inductor **1338** and the fourth inductor **1340**, and an anode connection coupled to the third terminal of the second relay **1334**, which is coupled to the energy storage device **1332**. The seventh diode **1336** enables a faster transition from a normal mode of operation to a backup mode of operation.

(85) For example, in one embodiment, the UPS **1300** may transition from a normal mode of operation to the backup mode of operation. Accordingly, the controller **1305** may control the second relay **1334** to transition from connecting the third terminal to the second terminal (which may be a configuration of the second relay **1334** during the normal mode of operation) to connecting the third terminal to the first terminal (which may be a configuration of the second relay **1334** during the backup mode of operation). Before the second relay **1334** completes the transition, current from the energy storage device **1332** may be unable to pass from the third terminal of the second relay **1334** to the first terminal of the second relay **1334**, because a conductive path through the second relay **1334** is not yet completed.

(86) In one example, the seventh diode **1336** being connected in parallel with the second relay **1334** enables the energy storage device **1332** to discharge through a conductive path including the energy storage device **1332**, the seventh diode **1336**, and at least one of the third inductor **1338** and the second switching device **1342**, or the fourth inductor **1340** and the third switching device **1344**, while the second relay **1334** completes a transition from connecting the third terminal to the second terminal to connecting the third terminal to the first terminal. Stated differently, because the seventh diode **1336** is connected in parallel with the second relay **1334**, the energy storage device **1332** is allowed to discharge through the seventh diode **1336** before the second relay **1334** is fully transitioned from a normal mode of operation configuration and a backup mode of operation configuration. Accordingly, the seventh diode **1336** enables the energy storage device **1332** to begin discharging quickly in a transition from a normal mode of operation to a backup mode of operation.

(87) FIG. **14** illustrates a schematic diagram of a UPS **1400** according to an embodiment. The UPS **1400** includes a dual converter **1402**, an inverter **1404**, and a controller **1405**. The dual converter **1402** is configured to be coupled to an AC input **1406**, and is coupled to the inverter **1404**. The inverter **1404** is coupled to the dual converter **1402**, and is configured to be coupled to a load **1408**.

(88) The UPS **1400** is similar to the UPS **200**. For example, the dual converter **1402**, the inverter **1404**, and the controller **1405** are similar to the dual converter **202**, the inverter **204**, and the controller **205**. The UPS **1400** further includes a relay **1410** configured as a single-pole double-throw switching device having a first terminal, a second terminal, and a third terminal. The third terminal is configured to be switchably connected to one of the first terminal and the second terminal. The first terminal is configured to be coupled to the AC input **1406**. The second terminal is coupled to the inverter **1404**. The third terminal is configured to be coupled to the load **1408**.

(89) The relay **1410** is configured to enable a bypass mode of operation in addition to the backup mode of operation and the normal mode of operation. In the bypass mode of operation, the third terminal of the relay **1410** is coupled to the first terminal of the relay **1410** such that AC power provided by the AC input **1406** bypasses the dual converter **1402** and the inverter **1404**. In the normal mode of operation or the backup mode of operation, the third terminal of the relay **1410** is coupled to the second terminal of the relay **1410** such that the AC input **1406** provides input power to the dual converter **1402** such that the UPS **1400** operates similarly to the UPS **200**.

(90) The controller **1405** may provide one or more control signals to the relay **1410** to control the switching position of the relay **1410**. For example, the controller **1405** may select the bypass mode of operation, thereby controlling the relay **1410** to connect the third terminal of the relay **1410** to the first terminal of the relay **1410**, responsive to determining that the AC power provided by the AC input **1406** is of a sufficiently high quality. Determining that the AC power is of the sufficiently high quality may include determining that a parameter of the AC power is within a threshold range.
(91) For example, the controller **1405** may determine that the voltage of the AC power is within 1V of an ideal 120V sinusoidal waveform and that the AC power is therefore of a sufficiently high quality to provide directly to the load **1408**. Otherwise, and continuing with the foregoing example, if the voltage of the AC power is not within 1V of an ideal 120V sinusoidal waveform, the controller **1405** may determine that the AC power should be processed by the dual converter **1402** and the inverter **1404** before being provided to the load **1408**. Accordingly, the controller **1405** may control the relay **1410** to connect the third terminal of the relay **1410** to the second terminal and select one of the backup mode of operation or the normal mode of operation.
(92) As discussed above, UPSs disclosed herein may be controlled by controllers including the controller **205**, the controller **1205**, the controller **1305**, and the controller **1405**. Using data stored in associated memory, the controllers may execute one or more instructions stored on one or more non-transitory computer-readable media that may result in manipulated data. In some examples, the controllers may include one or more processors or other types of controllers. In another example, the controllers include a Field-Programmable Gate Array (FPGA) controller.
(93) In yet another example, the controllers perform a portion of the functions disclosed herein on a processor and performs another portion using an Application-Specific Integrated Circuit (ASIC) tailored to perform particular operations. As illustrated by these examples, examples in accordance with the present invention may perform the operations described herein using many specific combinations of hardware and software and the invention is not limited to any particular combination of hardware and software components.
(94) As discussed above, the UPS **200** may receive power from the energy storage device **228**, which may be internal to, or external to and coupled to, the UPS **200**. In some embodiments, the energy storage device **228** may be coupled to an external charging device (not illustrated) configured to charge the energy storage device **228** with electrical energy. In alternate embodiments, the dual converter **202** may be configured to recharge the energy storage device **228** with energy derived from the AC power source **206**.
(95) For example, recharging the energy storage device **228** may include controlling the second relay **230** and at least one of the second switching device **236** and the third switching device **238** to provide a conductive path including the energy storage device **228**, the second relay **230**, and either one or both of the third inductor **232** and the second switching device **236**, and the fourth inductor **234** and the third switching device **238**. At least a portion of the power provided by the AC power source **206** may be provided to the energy storage device **228** to charge the energy storage device **228** via at least one of the first inductor **218** and the third inductor **232**, and the second inductor **220** and the fourth inductor **234**.
(96) In one example, charging the energy storage device **228** may include controlling the second relay **230** to connect the third terminal of the second relay **230** to the first terminal of the second relay **230** and controlling the second switching device **236** to be in a closed and conducting position during a positive half-cycle of AC power provided by the AC power source **206**. The first inductor **218** may be energized by the positive half-cycle of the AC power provided by the AC power source **206** and energize the magnetically coupled third inductor **232**. The third inductor **232** may discharge induced current to the energy storage device **228** through a conductive path including the energy storage device **228**, the second relay **230**, the third inductor **232**, and the second switching device **236**, thereby charging the energy storage device **228**.
(97) In another example, charging the energy storage device **228** may include controlling the

second relay **230** to connect the third terminal of the second relay **230** to the first terminal of the second relay **230** and controlling the third switching device **238** to be in a closed and conducting position during a negative half-cycle of AC power provided by the AC power source **206**. The second inductor **220** may be energized by the negative half-cycle of the AC power provided by the AC power source **206** and energize the magnetically coupled fourth inductor **234**. The fourth inductor **234** may discharge induced current to the energy storage device **228** through a conductive path including the energy storage device **228**, the second relay **230**, the fourth inductor **234**, and the third switching device **238**, thereby charging the energy storage device **228**.
(98) In light of the foregoing remarks, UPSs having reduced size and component counts with increased flexibility have been described herein. In one example, a UPS having a dual converter has been described. The dual converter includes components configured to operate during a backup mode of operation and a normal mode of operation, whereas certain conventional UPSs include a first set of components to operate during a backup mode of operation, and a second set of components to operate during a normal mode of operation. Furthermore, the dual converter is configured to control an amount of power provided to each individual DC bus capacitor, whereas certain conventional UPSs are only capable of providing an equal amount of power each DC bus capacitor. Accordingly, a component count may be reduced, thereby lowering a cost and physical size of the UPS, while increasing the flexibility of the UPS.
(99) Although certain embodiments have illustrated a dual converter implemented in connection with a UPS, other exemplary converters may not be implemented in connection with a UPS. The dual converter may be implemented in any other environment or topology, and is not limited to examples including UPSs. For example, the dual converter may be implemented with a power device other than a UPS.
(100) Having thus described several aspects of at least one embodiment of this invention, it is to be appreciated various alterations, modifications, and improvements will readily occur to those skilled in the art. Such alterations, modifications, and improvements are intended to be part of this disclosure, and are intended to be within the spirit and scope of the invention. Accordingly, the foregoing description and drawings are by way of example only.

## Claims

1. An uninterruptible power supply (UPS) system including: a first input configured to be coupled to an AC input power source and configured to receive AC input power having a positive cycle and a negative cycle from the AC input power source; a second input configured to be coupled to an energy storage device; an output configured to provide output power; a power conversion circuit configured to convert power received from at least one of the AC input power source or the energy storage device, the power conversion circuit including a primary branch portion having a first primary branch path including a first inductor and a second primary branch path coupled in parallel with the first primary branch path, and a backup branch portion having a second inductor, the second inductor being magnetically coupled to, and galvanically isolated from, the first inductor; an output circuit coupled to the power conversion circuit and to the output; and at least one controller coupled to the power conversion circuit and to the output circuit, and configured to control the power conversion circuit to provide, in a normal mode of operation and during the positive cycle of the AC input power, a first current derived from the AC input power source to the output circuit through the first inductor and not through the second primary branch path, wherein the first current through the first inductor induces a first induced voltage across the second inductor, discharge, by the second inductor in the normal mode of operation, a second current derived from the first induced voltage to the energy storage device, draw, in a backup mode of operation, a third current from the energy storage device through the second inductor, wherein the third current through the second inductor induces a second induced voltage across the first inductor,

and discharge, by the first inductor, a fourth current derived from the second induced voltage to the output circuit.

2. The UPS system of claim 1, wherein the output circuit includes a first capacitor configured to be coupled to the power conversion circuit and a second capacitor configured to be coupled to the power conversion circuit.

3. The UPS system of claim 1, wherein the second primary branch path of the primary branch portion includes a third inductor, and the backup branch portion includes a fourth inductor, and wherein the third inductor is magnetically coupled to the fourth inductor.

4. The UPS system of claim 3, wherein the second inductor is coupled to a first switching device, and wherein the fourth inductor is coupled to a second switching device.

5. The UPS system of claim 4, wherein the at least one controller is further configured to control the first switching device to induce a voltage across the first inductor and discharge the first inductor to charge a first capacitor, and is configured to control the second switching device to induce a voltage across the third inductor and discharge the third inductor to charge a second capacitor.

6. The UPS system of claim 5, wherein the at least one controller is further configured to control the first switching device to charge the first capacitor at a first rate and is configured to control the second switching device to charge the second capacitor at a second rate different than the first rate.

7. The UPS system of claim 5, wherein the at least one controller is further configured to control the first switching device and the second switching device to charge the first capacitor simultaneously with the second capacitor.

8. The UPS system of claim 5, wherein controlling the first switching device to charge the first capacitor includes: controlling the first switching device to enable the energy storage device to provide current to the second inductor, wherein providing current to the second inductor includes inducing a voltage across the first inductor; and controlling the first switching device to disable the energy storage device from providing current to the second inductor, wherein the first inductor is configured to discharge to the first capacitor responsive to the first switching device disabling the energy storage device.

9. The UPS system of claim 8, wherein controlling the second switching device to charge the second capacitor includes: controlling the second switching device to enable the energy storage device to provide current to the fourth inductor, wherein providing current to the fourth inductor includes inducing a voltage across the third inductor; and controlling the second switching device to disable the energy storage device from providing current to the fourth inductor, wherein the third inductor is configured to discharge to the second capacitor responsive to the second switching device disabling the energy storage device.

10. The UPS system of claim 1, wherein the backup branch portion further includes: a switching device having a first connection coupled to the energy storage device and a second connection switchably coupled to the second inductor; and a diode having an anode connection coupled to the first connection of the switching device and a cathode connection coupled to the second connection of the switching device.

11. The UPS system of claim 1, further comprising a switching device including: a first connection coupled to the output; and a second connection configured to be coupled to one of the output circuit or the first input, wherein the at least one controller is configured to control the switching device to connect the second connection to the first input in a bypass mode of operation, and is configured to control the switching device to connect the second connection to the output circuit in the normal mode of operation and the backup mode of operation.

12. The UPS system of claim 3, further comprising: a first diode having a cathode connection coupled to the first inductor, and an anode connection coupled to a reference node; and a second diode having a cathode connection coupled to the reference node, and an anode connection coupled to the third inductor.

13. A method of controlling a power device comprising a first input to receive first AC input power having a positive cycle and a negative cycle, a second input to receive second input power and configured to be coupled to an energy storage device, an output to provide output power, an output circuit to provide the output power to the output, and a power conversion circuit including a primary branch portion having a first primary branch path including a first inductor and a second primary branch path coupled in parallel with the first primary branch path, and a backup branch portion having a second inductor, the second inductor being magnetically coupled to, and galvanically isolated from, the first inductor, the method comprising: providing, in a normal mode of operation and during the positive cycle of the first AC input power, a first current derived from the first AC input power to the output circuit through the first inductor and not through the second primary branch path, wherein the first current through the first inductor induces a first induced voltage across the second inductor; discharging, by the second inductor in the normal mode of operation, a second current derived from the first induced voltage to the energy storage device; drawing, in a backup mode of operation, a third current from the energy storage device through the second inductor, wherein the third current through the second inductor induces a second induced voltage across the first inductor; and discharging, by the first inductor, a fourth current derived from the second induced voltage to the output circuit.

14. The method of claim 13, wherein the output circuit includes a first capacitor and a second capacitor, wherein the second primary branch path of the primary branch portion further includes a third inductor, and wherein the backup branch portion includes a fourth inductor magnetically coupled to the third inductor, the method further comprising providing, by the first inductor and the third inductor, a first charging current to the first capacitor and a second charging current to the second capacitor, respectively.

15. The method of claim 14, the method further comprising providing the first charging current to the first capacitor at a first rate and provide the second charging current to the second capacitor at a second rate, wherein the first rate is different than the second rate and wherein the first charging current is provided simultaneously with the second charging current.

16. The method of claim 13, wherein the power device further includes a switching device having a first connection coupled to the output, and a second connection configured to be coupled to one of the output circuit or the first input, the method further comprising connecting the second connection to the first input in a bypass mode of operation, and connecting the second connection to the output circuit in the normal mode of operation and the backup mode of operation.

17. A non-transitory computer-readable medium storing thereon sequences of computer-executable instructions for controlling a power device comprising a first input to receive first AC input power having a positive cycle and a negative cycle, a second input to receive second input power, an output to provide output power, an output circuit to provide the output power to the output, and a power conversion circuit including a primary branch portion having a first primary branch path including a first inductor and a second primary branch path coupled in parallel with the first primary branch path, and a backup branch portion having a second inductor, the second inductor being magnetically coupled to the first inductor, the sequences of computer-executable instructions including instructions that instruct at least one processor to control the power conversion circuit in the power device to: provide, to the first inductor and not through the second primary branch path in a normal mode of operation and during the positive cycle of the first AC input power, the first AC input power; provide, by the first inductor in the normal mode of operation, first power derived from the first AC input power to the output circuit; receive, by the second inductor in a backup mode of operation, the second input power; store, by the second inductor in the backup mode of operation, first stored energy derived from the second input power; store, by the first inductor in the backup mode of operation, second stored energy derived from the first stored energy; and provide, by the first inductor in the backup mode of operation, second power derived from the second stored energy to the output circuit.

18. The non-transitory computer-readable medium of claim 17, wherein the output circuit includes a first capacitor and a second capacitor, wherein the second primary branch path of the primary branch portion further includes a third inductor, and wherein the backup branch portion includes a fourth inductor magnetically coupled to the third inductor, the sequences of computer-executable instructions further including instructions that instruct the at least one processor to control the power conversion circuit to provide, by the first inductor and the third inductor, a first charging current to the first capacitor and a second charging current to the second capacitor.
19. The non-transitory computer-readable medium of claim 18, the sequences of computer-executable instructions further including instructions that instruct the at least one processor to control the power conversion circuit to provide the first charging current to the first capacitor at a first rate and provide the second charging current to the second capacitor at a second rate, wherein the first rate is different than the second rate and wherein the first charging current is provided simultaneously with the second charging current.
20. The non-transitory computer-readable medium of claim 17, wherein the power device further includes a switching device having a first connection coupled to the output, and a second connection configured to be coupled to one of the output circuit or the first input, and wherein the sequences of computer-executable instructions further include instructions that instruct the at least one processor to control the switching device to connect the second connection to the first input in a bypass mode of operation, and to control the switching device to connect the second connection to the output circuit in the normal mode of operation and the backup mode of operation.